

S Madhan Kumar

A highly skilled Data Analyst and Lead Developer with a Bachelor's in Computer Science specializing in Data Science. Proficient in developing end-to-end analytical pipelines, optimizing workflows, and synthesizing critical business intelligence. Experienced in leveraging Python, Java, SQL, and various AI/ML frameworks to build scalable solutions, including a patent-pending framework for environmental analytics and AI-powered character recognition systems. Demonstrates expertise in data visualization, predictive modeling, and full-stack development to drive strategic initiatives.

✉ smk06056@gmail.com 📞 +91 9789823776 📍 Chennai, Tamil Nadu in [\[LINK\] linkedin.com/in/madhan-kumar-ds](#) gh [\[LINK\] github.com/madhan1012](#)

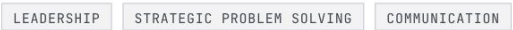
[02] Technical_Engine



[03] Technical_Skills



[04] Soft_Skills



[05] Experience_Log

Jan 2025 - Feb 2025

● ACTIVE

Data Science Intern

Ediglobe

[REF: EX-001]

- Engineered end-to-end analytical pipelines to solve complex business bottlenecks, contributing to quarterly strategic goal achievement.
- Optimized operational workflows by leveraging data insights, resulting in measurable enhancements to business value and process efficiency.
- Synthesized critical business intelligence for high-level stakeholders, facilitating data-driven decision-making through advanced statistical modeling.
- Developed interactive data visualizations and predictive models to interpret trends and influence strategic company-wide initiatives.

[06] Project_Schematics

PROJ_001 [\[LINK\] github.com/Madhan1012/VeIdora](#)

Multi-Modal MRV Framework

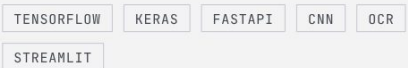
Developed a patent-pending hybrid framework for Soil Organic Carbon (SOC) estimation using IoT and Satellite data. This system fuses Sentinel-2 multi-spectral imagery with real-time ground telemetry from ESP32 sensors, along with automated data ingestion pipelines for high-precision environmental analytics.



PROJ_002 [\[LINK\] github.com/Madhan1012/Character-Recognition-V2](#)

Ancient Tamil Character Recognition

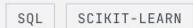
Built an AI-powered web application using TensorFlow/Keras and FastAPI to decipher 8th-century Vatteluthu inscriptions. This involved developing a CNN-based OCR model to translate ancient scripts into modern Tamil, preserving historical linguistic data, and featuring a Streamlit frontend.



PROJ_003 [\[LINK\] github.com/Madhan1012/Natural-Language-To-SQL](#)

NL-to-SQL Intent Engine

Created a modular bot designed to translate natural language queries into executable SQL commands against relational datasets. This engine implemented an intent-based classification system using Scikit-learn to reduce data access friction for non-technical users.



PROJ_004 [\[LINK\] github.com/Madhan1012/AegisLM](#)

AegisLM

Architected a Template-LM hybrid system utilizing Java Spring Boot and Python to streamline local LLM orchestration. Engineered a complex relational database schema in MySQL with recursive joins and designed integration protocols for local inference servers, including LM Studio and Qwen 2.5 Coder.

- JAVASPRING BOOTPYTHON
- LLM ORCHESTRATIONMYSQL
- RECURSIVE JOINSLM STUDIO
- QWEN 2.5 CODER

[07]

Education

Bachelor’s of Technology in ComputerScience (Data Science)

VeL Tech University

2023 - Present

12th in MPCB

Everwin Vidhyasharam

2022 - 2023